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Editorial board

Editors for the Journal of Mathematical Imaging & Vision,

Enclosed please find for consideration our original research article entitled:

Explainable Binary Classification of Separable Shape Ensembles

by Z. Grey, N. Fisher, and A. Glaws. We confirm that the manuscript has not been published previously, is not under review by any other journal, and that all authors have approved its submission to Journal of Mathematical Imaging and Vision (JMIV).

Image segmentation is a cornerstone of modern scientific imaging. In disciplines ranging from materials science to cellular biology, practitioners routinely extract thousands of segmented curves that encode the morphology of structures of interest. These curve ensembles are subsequently used to detect subtle phenotypic changes, quantify growth patterns, or monitor disease progression.

Despite the ubiquity of large curve ensembles, existing shape analysis pipelines face three fundamental challenges:

Scalability. Existing AI models capable of insightful comparison between two images require training over hand-labeled data, constantly changing selection of opaque model forms, and are remarkably expensive to train and operate.

Shape representation. Rigid motions (translation, rotation, reflection) and nonlinear undulations are intertwined in most representations, hindering interpretability. Classical representations (Fourier descriptors) do not offer insight into simple, fundamental distinguishing features of shape.

Explainability. Black-box classifiers (deep networks, kernel SVMs over alternative feature spaces) can achieve high accuracy but provide little insight into which geometric variations drive the decision and are incredibly expensive to train.

Our manuscript directly addresses these gaps by introducing a mathematically rigorous, computationally efficient framework that transforms massive curve ensembles into dual, separable shape tensors representing interpretable decomposition into linear and nonlinear constituents of shape deformation. This is achieved in an entirely 'unsupervised' manner—i.e., labeled data is not required and decisions are made intrinsically based on the feature distributions of the paired ensembles.

The paper presents four specific innovations among others:

Highly Accurate Composite Integral Operator Approximation. We expound on a simple compact integral operator that can be used to encode the geometry of all curves in an ensemble. By accurately approximating its eigenspaces using a quadrature-based Nyström method, we obtain a discrete representation that lives on a set of quadrature nodes common to every curve according to a fixed integral measure.

Separable Shape Tensor (SST) Decomposition. The resulting quadrature admits a rigid-invariant decomposition into a generalized scale mode (capturing linear size variations) and a nonlinear undulation mode (capturing nonlinear shape-specific deformations). This decomposition is shown to be mathematically analogous to a polar decomposition, guaranteeing uniqueness up to orthogonal group actions.

Dual Curve Representation on Matrix Manifolds. We present a novel analysis naturally identified SST approximations with elements on the Stiefel manifold, yielding dual curve representations that are simultaneously amenable to linear algebraic manipulation and the geometric constraints of the underlying shape space. Then, the rigid invariance of the transformed discrete curves is subsequently retained by projection onto the underlying Grassmannian manifold.

Explainable Binary Classification via Product MMD. By learning normal coordinate distributions and embedding these distributions in a product Maximum Mean Discrepancy (MMD) space, we construct a lightweight, data-driven classifier whose decision function is serves as a logical conjunction between generalized scales and nonlinear undulations of curves. Crucially, the classifier can be trained without labeled data, exploiting the MMD's two-sample test interpretation, and the resulting geodesic distances emphasize novel visualization and approximations of data.

Together, these contributions enable the interactive analysis of thousands of segmented curves from an image (or set of images) in a matter of seconds on a standard workstation, a speedup of several orders of magnitude compared with existing shape analysis pipelines. *Additionally, we demonstrate how the method is capable of making intuitive inferences on scales well below human visual scales.*

JMIV is uniquely positioned at the intersection of rigorous mathematical modeling and practical image analysis. Our work leverages functional analysis (integral operators), differential geometry (matrix manifolds), and statistical learning (MMD)—areas that are core to the journal's scope. Moreover, the manuscript introduces a new analytical pipeline that can be readily incorporated into existing imaging workflows, thereby appealing to both theoretical readers and experimentalists needing rapid, scalable, interpretable shape analytics.

We would be grateful for the opportunity to contribute this work to Journal of Mathematical Imaging and Vision. Thank you for your careful consideration.

Sincerely,

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